



Singapore University of Technology and Design

60.001 Applied Deep Learning Project

**Topic: Deep Learning for Early Alzheimer's Diagnosis via Structural MRI
Volumetric Analysis**

Github Link: <https://github.com/Xinpingxp/ARIA.git>

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1. Introduction

1.1 Background and Motivation

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia worldwide, accounting for an estimated 60–80% of all dementia cases. According to the World Health Organization (WHO), over 55 million people currently live with dementia globally, with nearly 10 million new cases diagnosed each year. As populations age, this figure is projected to rise to 139 million by 2050, placing an enormous and growing burden on healthcare systems, caregivers, and economies worldwide.

Despite its prevalence, Alzheimer's Disease remains critically underdiagnosed in its earliest and most treatable stages. The disease follows a long preclinical trajectory which includes the pathological changes in the brain, including the accumulation of amyloid-beta plaques and tau neurofibrillary tangles, begin silently decades before any clinical symptoms emerge. The transitional stage between normal cognitive aging and clinically diagnosed AD is known as Mild Cognitive Impairment (MCI). Individuals at this stage exhibit subtle cognitive decline that does not yet significantly impair daily functioning, yet up to 15% of MCI patients convert to full Alzheimer's dementia annually. Early identification of MCI patients at high conversion risk is therefore a critical clinical priority.

Structural Magnetic Resonance Imaging (sMRI) has become one of the most widely adopted neuroimaging modalities in both clinical practice and AD research. It offers a non-invasive, radiation-free means of visualising brain morphology in high spatial resolution. Characteristic patterns of brain atrophy, particularly in the hippocampus, entorhinal cortex, medial temporal lobe, and posterior cingulate cortex which have been consistently associated with AD pathology and cognitive decline. These regions are among the earliest affected by neurodegeneration and serve as established biomarkers for disease progression.

However, the manual clinical interpretation of sMRI scans for early-stage atrophy presents significant challenges. In routine clinical settings, radiologists and neurologists rely on visual inspection, which is inherently subjective, time-intensive, and prone to inter-observer variability. The volumetric changes at the MCI stage are often subtle, sometimes amounting to only a few percent reduction in regional brain volume, and fall well within the range of normal individual anatomical variation. This makes reliable early detection exceptionally difficult without quantitative, automated tools.

The past decade has witnessed remarkable advances in deep learning, particularly in the domain of medical image analysis. Convolutional Neural Networks (CNNs) and more recently Vision Transformers (ViTs) have demonstrated expert-level or superior performance across a range of tasks including tumour segmentation, retinal disease detection, and radiological classification. In

the context of neuroimaging, deep learning models have shown considerable promise in automating whole-brain segmentation, extracting regional volumetric features, and classifying diagnostic groups such as AD versus healthy controls. Yet a significant gap remains: the automated, reliable classification of the MCI-to-AD conversion risk at scale, the exact clinical scenario where early intervention would be most impactful; it remains an open and challenging research problem.

This project is therefore motivated by the convergence of three pressing realities: the growing global burden of Alzheimer's Disease, the clinical inadequacy of manual MRI interpretation for early-stage detection, and the demonstrated potential of deep learning to extract quantitative, reproducible biomarkers from neuroimaging data that exceed human visual sensitivity.

1.2 Problem Statement

Despite the well-established role of structural MRI in neuroimaging, current clinical workflows lack a reliable and scalable mechanism for objectively quantifying the subtle brain atrophy changes that characterise the early Mild Cognitive Impairment (MCI) phase of Alzheimer's Disease. Manual radiological review, while valuable, is inherently limited in its sensitivity to the marginal volumetric reductions particularly in the hippocampus, entorhinal cortex, and medial temporal lobe that distinguish early AD-driven neurodegeneration from normal age-related anatomical variation. This limitation introduces significant inter-observer variability and reduces diagnostic consistency across clinical settings.

The consequences of this gap are directly clinical. MCI patients who are not accurately risk-stratified miss the opportunity for early therapeutic intervention, structured care planning, and eligibility for disease-modifying clinical trials, all of which are considerably more impactful when initiated before significant cognitive decline has occurred. At the same time, the absence of objective quantification tools means that clinicians must make high-stakes prognostic decisions based on subjective visual assessment alone, without a standardised, data-driven reference.

This project therefore addresses a specific and unmet need: the development of an automated deep learning pipeline that quantifies regional brain atrophy from structural MRI scans to predict AD conversion risk at the MCI stage. Rather than replacing clinical judgement, the system is designed to augment it, providing reproducible, quantitative atrophy measurements across neuroanatomically relevant regions and translating these into a patient-level risk stratification across four diagnostic categories: cognitively normal (CN), stable MCI (sMCI), progressive MCI (pMCI), and Alzheimer's Disease (AD). The core technical challenge lies in building a model that is sensitive enough to detect subtle, early-stage morphometric changes while remaining generalisable across the natural variation present in real-world populations.

2. Dataset & Collection

2.1 Dataset Overview

The dataset used in this project is the OASIS Alzheimer's Detection dataset, sourced from the Kaggle repository (ninadaithal/imagesoasis) and originally derived from the Open Access Series of Imaging Studies (OASIS), which is a neuroimaging initiative led by Washington University in St. Louis with the explicit goal of making brain MRI data freely and openly available to the broader scientific and research community.

The underlying OASIS-1 source collection is a cross-sectional dataset comprising 416 subjects aged 18 to 96, with each subject contributing 3 to 4 individual T1-weighted MRI scans acquired within a single imaging session. All subjects were right-handed and included both males and females, with 100 subjects over the age of 60 having been clinically diagnosed with very mild to moderate Alzheimer's Disease.

For the Kaggle version used in this project, the raw volumetric MRI scans were preprocessed and converted into 2D image slices to facilitate deep learning model training. The original `.img` and `.hdr` files were first transformed into NIfTI format (`.nii`) using FSL, after which each 3D brain volume was divided into 256 axial segments. Slices 100 to 160 were selected from each as the most diagnostically relevant cross-sections, and CDR values alongside metadata were used to classify patients into four diagnostic groups.

The resulting dataset comprises approximately 80,000 preprocessed MRI images organised into four dementia severity classes: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. The images are structured into separate class-labelled folders, making them directly compatible with standard supervised deep learning classification pipelines. The dataset is publicly accessible, requires no institutional data access agreement, and is well-established in the Alzheimer's neuroimaging research literature, making it a practical and reproducible benchmark for this project.

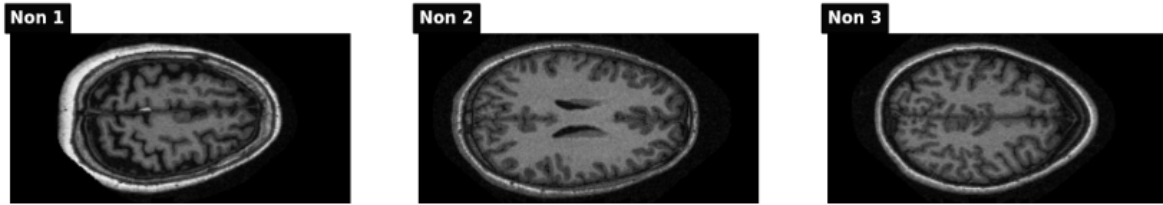


Figure 1: Images from Non-Demented Folder

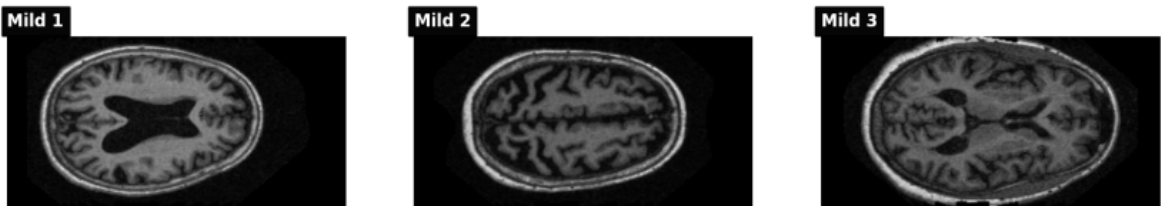


Figure 2: Images from Mild Demented Folder

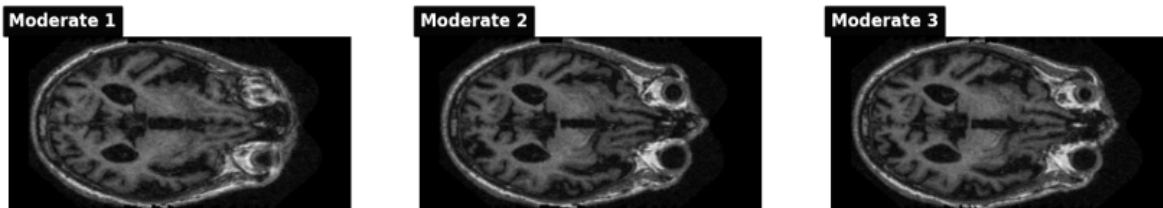


Figure 3: Images from Moderate Demented Folder

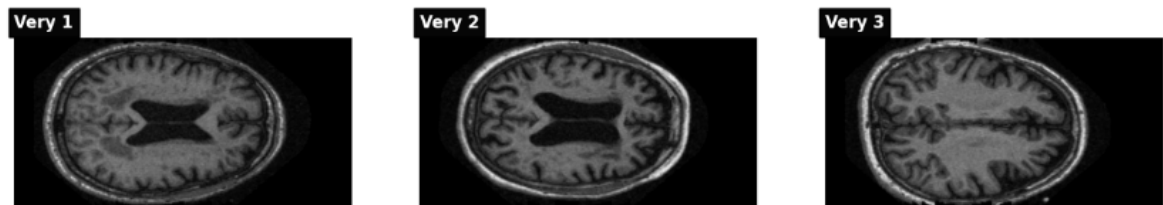


Figure 4: Images from Very Mild Demented Folder

2.2 Dataset Characteristics

2.2.1 Source and Origin

The dataset is derived from the OASIS-1 cross-sectional MRI study conducted at Washington University in St. Louis, one of the most widely cited open neuroimaging repositories in Alzheimer's research. The original volumetric scans were preprocessed and restructured for deep learning use, and are publicly hosted on Kaggle under the dataset identifier *imagesoasis*. Its open-access nature, clinical validity, and widespread use in published literature make it a well-suited benchmark for this project.

2.2.2 Imaging Modality and Acquisition Parameters

All images in the dataset were acquired using T1-weighted structural Magnetic Resonance Imaging (sMRI), the standard modality for assessing brain morphology and regional atrophy in Alzheimer's research. Scans were collected using both 1.5T and 3T MRI scanners, with acquisition parameters comprising a repetition time of 2,400 ms, an echo time of 2.14 ms, and a field of view of 256 mm $\times$ 256 mm. T1-weighted sequences were selected for their superior grey-white matter contrast and their established sensitivity to the volumetric atrophy patterns characteristic of neurodegeneration, particularly in the hippocampus and medial temporal lobe.

2.2.3 Image Format and Dimensions

The raw MRI volumes, originally stored in NIfTI (*.nii*) format, were converted into JPEG (*.jpg*) format to facilitate compatibility with standard deep learning image classification frameworks. Each image is represented in grayscale with a spatial resolution of 176 $\times$ 208 pixels. While the conversion from volumetric NIfTI to 2D JPEG slices introduces a reduction in three-dimensional spatial context, it substantially lowers computational overhead and enables the direct application of well-established 2D convolutional neural network architectures.

2.2.4 Total Dataset Size and Subject Coverage

The dataset comprises approximately 80,000 two-dimensional axial MRI slice images sourced from 461 patients. Each contributes multiple slices extracted from their full volumetric scan, meaning the total image count significantly exceeds the number of unique subjects. This distinction is important when designing train/validation/test splits, as slices from the same must be kept within the same partition to prevent data leakage and ensure that model performance

metrics reflect genuine generalisation to unseen subjects rather than unseen slices from already-seen patients.

2.2.5 Class Distribution and Diagnostic Labelling

The dataset is organised into four diagnostic classes corresponding to progressive stages of Alzheimer's Disease severity, with class labels assigned based on each patient's Clinical Dementia Rating (CDR) score, a standardised and clinically validated instrument used to quantify dementia severity. A CDR of 0 denotes a cognitively normal individual with no dementia, 0.5 indicates very mild cognitive impairment, 1 corresponds to mild dementia, and 2 corresponds to moderate dementia. These scores were derived from structured clinical assessments and cognitive evaluations conducted at the time of imaging, providing a clinically grounded ground truth for supervised model training. The four resulting classes and their respective image counts are as follows: Non-Demented (67,200 images), Very Mild Demented (13,700 images), Mild Demented (5,200 images), and Moderate Demented (488 images).

2.2.6 Class Imbalance

The dataset exhibits a pronounced class imbalance, with the Non-Demented class accounting for approximately 84% of all images. The Moderate Demented class, at only 488 images, represents less than 1% of the total dataset. This severe skew poses a significant risk during model training, as an unweighted classifier may achieve high overall accuracy simply by predicting the majority class, while performing poorly on the clinically critical minority classes — precisely those representing patients most in need of early identification. To mitigate this, a combination of strategies will be applied, including class-weighted loss functions to penalise misclassification of minority classes more heavily, targeted data augmentation to synthetically expand underrepresented classes, and the use of stratified sampling during dataset splitting to ensure proportional class representation across all partitions. Model evaluation will also rely on metrics beyond overall accuracy, such as per-class F1-score, sensitivity, specificity, and the area under the receiver operating characteristic curve (AUC-ROC), which provide a more faithful representation of performance across imbalanced classes.

2.2.7 Subject Demographics

Images were acquired from approximately 1,000 individuals, with a gender distribution of roughly 55% female and 45% male, and participant ages ranging from 42 to 90 years. This demographic spread reflects a realistic representation of the older adult population at risk for Alzheimer's Disease and lends the dataset a degree of biological diversity that supports model generalisation. However, the demographic metadata associated with individual slices is not directly embedded in the image filenames or folder structure, which limits the ability to perform subgroup-level performance analysis without additional preprocessing of the accompanying clinical metadata files.

2.2.8 Slice Selection

From each patient's full 256-slice axial volumetric scan, slices 100 to 160 were selected for inclusion in the dataset. This range corresponds to the mid-axial planes of the brain, which are most likely to capture the hippocampus, entorhinal cortex, and surrounding medial temporal lobe structures, the regions exhibiting the earliest and most diagnostically significant patterns of atrophy in Alzheimer's Disease. Restricting the dataset to this slice range reduces noise from non-informative superior and inferior brain regions while concentrating model learning on the anatomical areas most relevant to the classification task.

3. Data Preprocessing

3.1. Train, Validation, and Test Split Strategy

Prior to model training, a standardised preprocessing pipeline was applied to all images to ensure consistency and optimise input quality for deep learning. Images were resized to a uniform spatial resolution of 224×224 pixels to satisfy the fixed input dimensions required by ResNet18. Pixel intensity values were normalised using a fixed standard scheme with mean = 0.5 and standard deviation = 0.5 applied globally across all images. This fixed normalisation scheme provides stable and reproducible preprocessing without requiring computation of dataset-specific statistics, and is consistent with common practice in transfer learning where ImageNet-pretrained networks expect roughly zero-centred inputs.

Data augmentation was applied exclusively to the training set and included random horizontal flipping, small rotational perturbations (± 10 degrees), and brightness and contrast jitter (up to 0.2), all constrained to transformations that preserve the clinical integrity of the MRI slices

without introducing anatomically implausible distortions. No augmentation was applied to validation or test sets, which received only resizing and fixed normalisation.

The raw OASIS dataset contains four folder-based classes: Non Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia. These were remapped to three clinical labels: Cognitively Normal (CN, label 0), Mild Cognitive Impairment (MCI, label 1), and Alzheimer's Disease (AD, label 2), with both Mild and Moderate Dementia folders merged into the AD class.

Splits were performed at the slice level using stratified sampling (scikit-learn StratifiedShuffleSplit) to ensure each split preserves the original class distribution. The dataset was divided into 70% training, 20% validation, and 10% test sets. It should be noted that because multiple 2D slices are extracted from the same MRI scan, slices from the same patient may appear in more than one split. This is a known limitation of slice-level splitting and means reported accuracy metrics may be optimistic relative to a strict patient-level evaluation. This is discussed further in Section 7.1.

3.2. Handling Class Imbalance

The OASIS dataset presents a severe class imbalance that required careful consideration during training. After remapping to three classes, the training set reflects the following approximate distribution: Cognitively Normal (CN) accounts for roughly 77.8% of slices, Mild Cognitive Impairment (MCI) accounts for roughly 15.9%, and Alzheimer's Disease (AD) accounts for roughly 6.4%. Left unaddressed, this skew would cause a standard classifier to achieve high overall accuracy simply by predicting the majority class, making the model clinically useless for the minority classes that matter most.

Several strategies were considered and evaluated to address this imbalance.

3.2.1 Weighted Cross-Entropy Loss

The primary strategy adopted was a class-weighted cross-entropy loss function. Rather than treating each misclassification equally, the loss function assigns a higher penalty to errors on minority class samples. The per-class weights were computed as the inverse of each class's frequency in the training set, then normalised so that the weights sum to the number of classes (3). This means that a misclassification of an AD sample contributes proportionally more to the loss than a misclassification of a CN sample, pushing the model to pay more attention to the underrepresented classes during training. In PyTorch, this was implemented by passing the computed weight tensor directly to `nn.CrossEntropyLoss`.

3.2.2 WeightedRandomSampler

A WeightedRandomSampler was also evaluated as an alternative approach, where each training batch is constructed by oversampling minority class images and undersampling majority class images so that batches are more balanced. However, this strategy was found to reduce overall classification accuracy in practice, likely because the aggressive oversampling of the small AD class introduced repetitive samples that added noise to the gradient signal rather than new information. This approach was therefore not used in the final model.

3.2.3 Data Augmentation on Training Set

To increase the effective diversity of training samples without artificially replicating identical images, data augmentation was applied exclusively to the training set. The augmentation pipeline consists of random horizontal flipping, random rotation within plus or minus 10 degrees, and colour jitter with brightness and contrast variation of up to 0.2. These transformations were chosen specifically because they do not corrupt the anatomical structure of the MRI slices. Flipping and small rotations are reasonable for brain MRI as they simulate natural variation in positioning, while brightness and contrast jitter reflects the scanner-level intensity variation present in real clinical data. No augmentation was applied to validation or test sets, which received only resizing and normalisation.

3.2.4 Stratified Splits

To ensure that all three classes are proportionally represented in every partition, stratified sampling was applied when creating the train, validation, and test splits. This was implemented using scikit-learn's StratifiedShuffleSplit, which guarantees that the class distribution of the original dataset is preserved in each of the three splits. Without stratification, random splitting on an imbalanced dataset risks placing very few or zero AD samples in the test set, which would make evaluation meaningless.

3.2.5 Final Strategy

The final model uses a combination of weighted cross-entropy loss, training augmentation, and stratified splitting. No undersampling or oversampling of the actual image data was performed.

Evaluation relies on per-class metrics including precision, recall, F1-score, and AUC-ROC rather than overall accuracy alone, since overall accuracy on an imbalanced dataset can be misleading.

4. Modelling

4.1 Ordinal vs. Regular Classification

The disease severity labels in this dataset follow a natural clinical order: Cognitively Normal represents the absence of disease, Mild Cognitive Impairment represents an intermediate transitional stage, and Alzheimer's Disease represents confirmed neurodegeneration. This ordering is clinically meaningful because the degree of error matters: misclassifying a CN as AD is more consequential than misclassifying them as MCI, and vice versa.

Ordinal classification approaches were explored to capture this structure explicitly. These included custom ordinal loss functions and the CORAL (Consistent Rank Logits) framework, which reformulates multiclass ordinal classification as a set of binary threshold subtasks with shared feature representations. However, these approaches either introduced training instability or resulted in lower classification performance on the validation set compared to standard multiclass classification.

The final model uses standard multiclass classification with three integer class labels: CN (0), MCI (1), and AD (2). The loss function is cross-entropy, weighted by inverse class frequency to handle imbalance. While the model does not explicitly encode ordinal constraints, the integer label ordering and the weighted loss implicitly encourage the model to distinguish adjacent severity levels. This approach achieved the best validation accuracy in practice and was selected as the final configuration.

4.2 Model Architecture

4.2.1 Backbone: ResNet18 with ImageNet Pretraining

The backbone of the ARIA classification system is a ResNet18 convolutional neural network, initialised with weights pretrained on the ImageNet dataset using PyTorch's ResNet18_Weights.IMAGENET1K_V1. ResNet18 belongs to the Residual Network family introduced by He et al. (2016), which solved the vanishing gradient problem in deep networks by introducing skip connections, also called residual connections, that allow gradients to bypass one or more layers during backpropagation. This makes it possible to train deeper networks more stably without degradation in performance.

ResNet18 contains 18 layers and approximately 11 million trainable parameters. It is one of the most widely used architectures in transfer learning for medical imaging tasks due to its strong pretrained feature representations, moderate computational cost, and well-understood behaviour. While larger architectures such as ResNet50 or DenseNet121 exist, ResNet18 was selected as a pragmatic choice that balances representation capacity with the risk of overfitting on a single-site MRI dataset.

4.2.2 Modifications for Grayscale MRI Input

The original ResNet18 expects 3-channel RGB images as input, reflecting its ImageNet pretraining. Since MRI scans are single-channel grayscale images, the first convolutional layer was replaced with a new Conv2d layer accepting 1 input channel instead of 3, while keeping all other parameters identical: kernel size 7x7, stride 2, padding 3, no bias. The remaining pretrained weights were retained and fine-tuned during training.

4.2.3 Classification Head

The original fully connected output layer of ResNet18 was replaced with a custom two-layer head consisting of a Dropout layer with dropout probability $p=0.5$, followed by a Linear layer mapping 512 features to 3 output classes (CN, MCI, AD). The Dropout layer acts as a regulariser during training by randomly zeroing activations, which reduces co-adaptation between neurons and mitigates overfitting. The final Linear layer produces unnormalised logits, which are passed to the weighted cross-entropy loss during training and to a softmax function during inference to produce class probabilities.

4.2.4 Full Architecture Summary

Component	Details
Backbone	ResNet18 (pretrained, ImageNet)
Input	1-channel grayscale, 224x224 pixels
Residual Blocks	4 stages (layer1 to layer4)
Feature Dimension	512 (after global average pooling)
Classification Head	Dropout(p=0.5) + Linear(512, 3)
Output Classes	3 (CN, MCI, AD)
Total Parameters	~11 million

Table 1: Table of our Model Architecture

4.2.5 Grad-CAM Integration

To support clinical interpretability, the model includes a `get_feature_maps()` method that exposes the output of the final residual block (layer4) for use with Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM computes a weighted combination of the feature maps from the last convolutional layer to produce a heatmap highlighting which spatial regions of the input image most influenced the model's classification decision. This is surfaced in the ARIA web interface so that clinicians can visually inspect which areas of the brain scan drove the prediction.

4.3 Hyper Parameter Tuning

Hyperparameters were selected based on standard practices for transfer learning with pretrained CNNs on medical imaging tasks, with adjustments made based on validation performance across training runs.

4.3.1 Final Hyperparameter Configuration

Hyperparameter	Value	Justification
Learning rate	1e-3	Standard starting point for Adam with pretrained backbone
Optimizer	Adam	Adaptive learning rate, well-suited to sparse and noisy gradients
Weight decay (L2)	1e-4	Prevents overfitting by penalising large weight magnitudes
Batch size	32	Balances memory usage and gradient estimate quality
Dropout rate	0.5	Strong regularisation for a relatively small domain-specific dataset
Epochs	5	Sufficient for convergence; early stopping active with patience 5
LR scheduler	ReduceLROnPlateau	Halves learning rate when val_loss does not improve for 3 epochs
LR reduction factor	0.5	Halves the learning rate on plateau
LR scheduler patience	3 epochs	Allows temporary val_loss fluctuation before reducing LR
Early stopping patience	5 epochs	Stops training if val_loss does not improve
Image size	224x224	Standard input size for ImageNet-pretrained ResNet
Normalisation	mean=0.5, std=0.5	Applied to single grayscale channel
Class weights	Inverse frequency, normalised to sum to 3	Handles severe class imbalance

Table 2: Table of our Selected Hyperparameters

4.3.2 Learning Rate and Optimiser

Adam was chosen over SGD because it adapts the learning rate individually for each parameter based on first and second moment estimates of gradients. This makes it more robust to sparse updates, which is beneficial when fine-tuning a pretrained network where different layers may need different effective learning rates. A learning rate of 1e-3 was used as the initial value, which is a well-established default for Adam in transfer learning settings.

4.3.3 L2 Regularisation

Weight decay of $1e-4$ was applied through the Adam optimiser's `weight_decay` parameter. This adds a penalty proportional to the squared magnitude of each weight to the total loss, discouraging the model from relying on any single large weight and improving generalisation to unseen data. This was particularly important given the relatively small number of AD samples in the dataset.

4.3.4 Dropout

A Dropout layer with $p=0.5$ was placed immediately before the final linear classifier. This randomly disables 50% of neurons during each training forward pass, preventing the network from memorising specific training patterns and forcing it to learn more redundant and generalisable representations. Dropout is disabled automatically during inference.

4.3.5 Learning Rate Scheduling

A `ReduceLROnPlateau` scheduler was configured to monitor validation loss. If validation loss does not decrease for 3 consecutive epochs, the learning rate is multiplied by 0.5. This allows the model to make large updates early in training and progressively refine with smaller updates as it converges, helping escape local minima while avoiding overshooting near the optimum.

4.3.6 Early Stopping

Early stopping was applied with a patience of 5 epochs monitoring validation loss. This prevents the model from continuing to train after it has stopped generalising, avoiding overfitting to the training set. The best model checkpoint was saved based on the highest validation accuracy observed during training.

5. User Interface

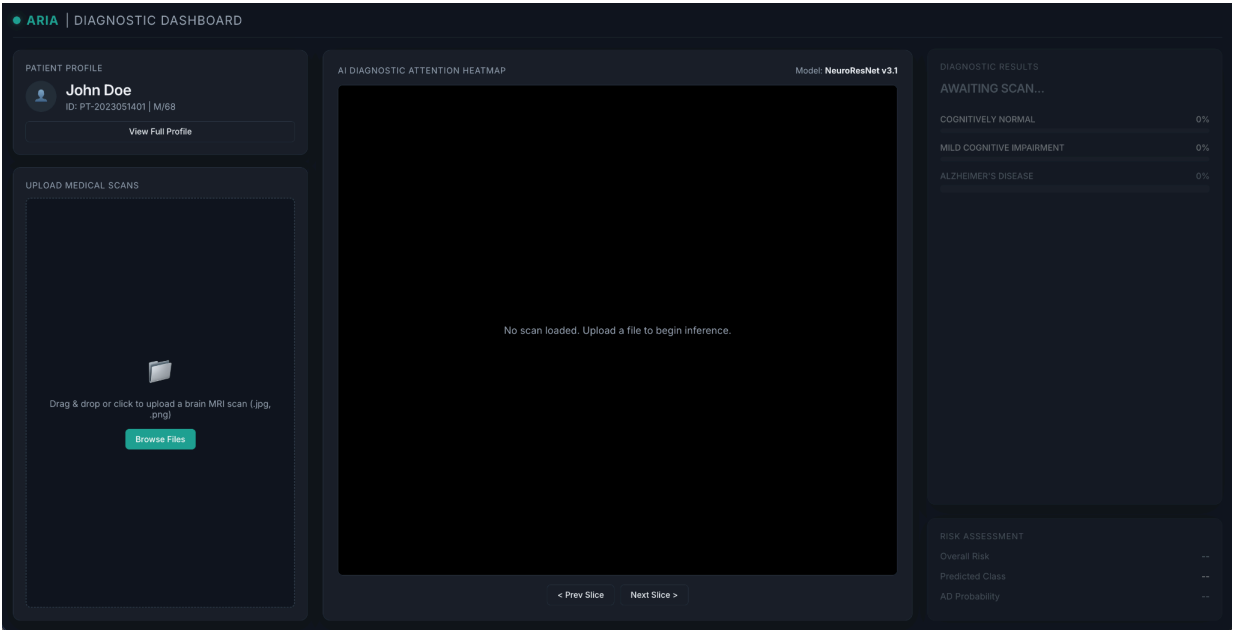


Figure 6: Dashboard UI of ARIA

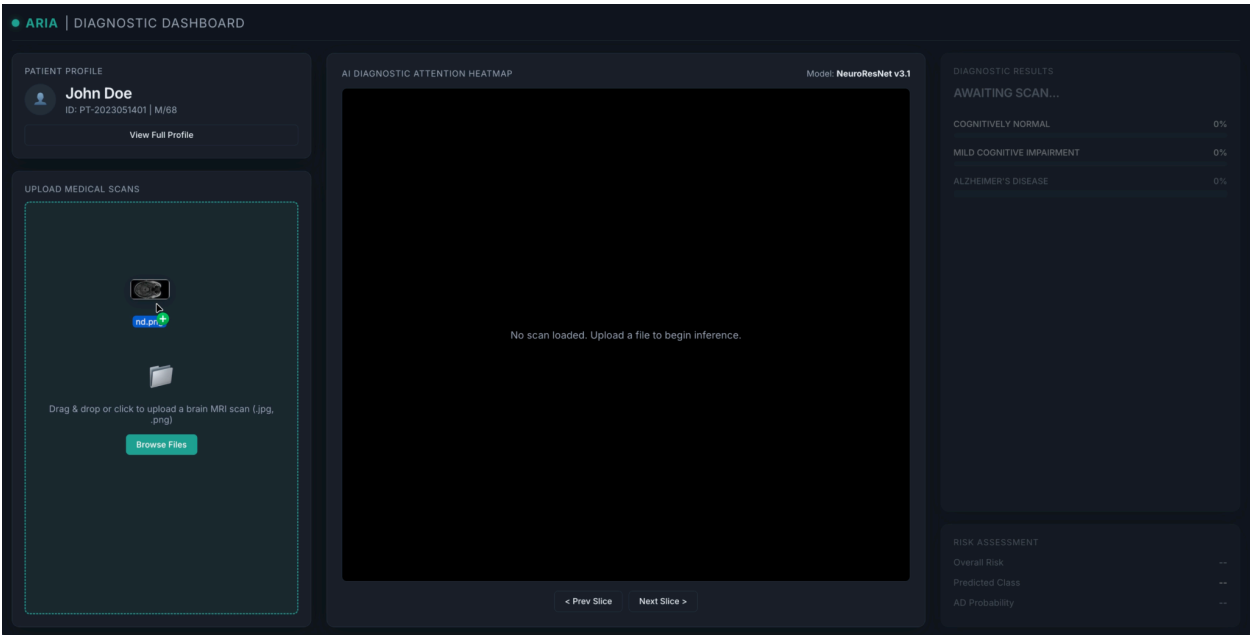


Figure 7: Uploading of an Image

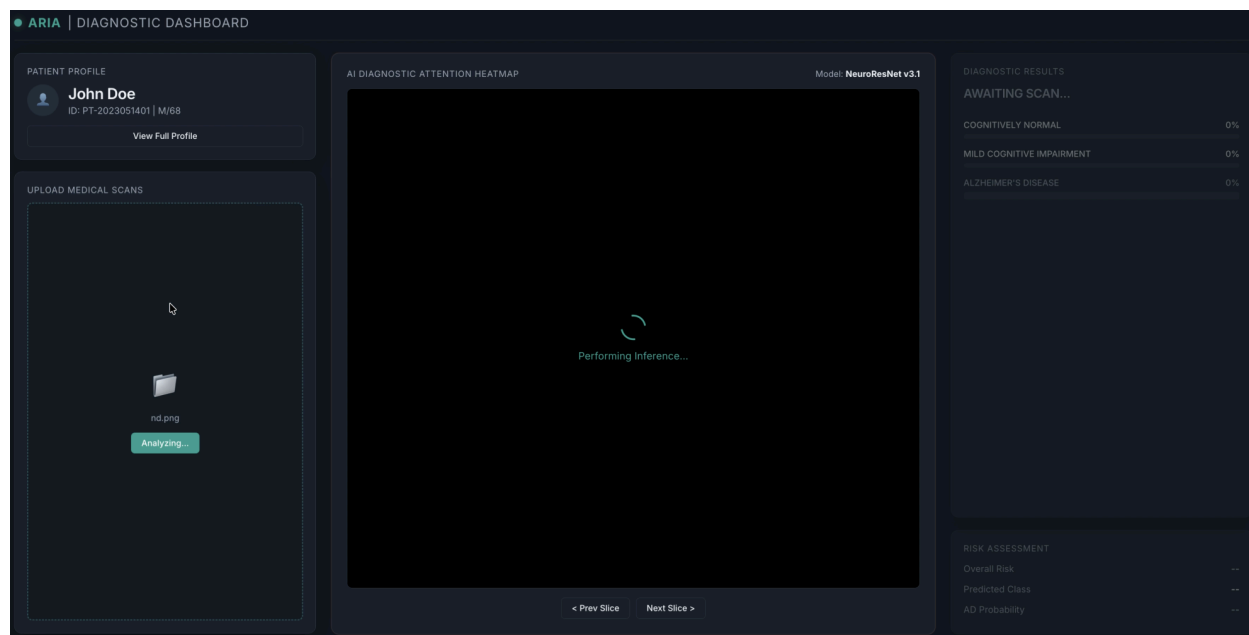


Figure 8: Analysis of MRI Medical Scans (eg image was a non-demented medical scan)

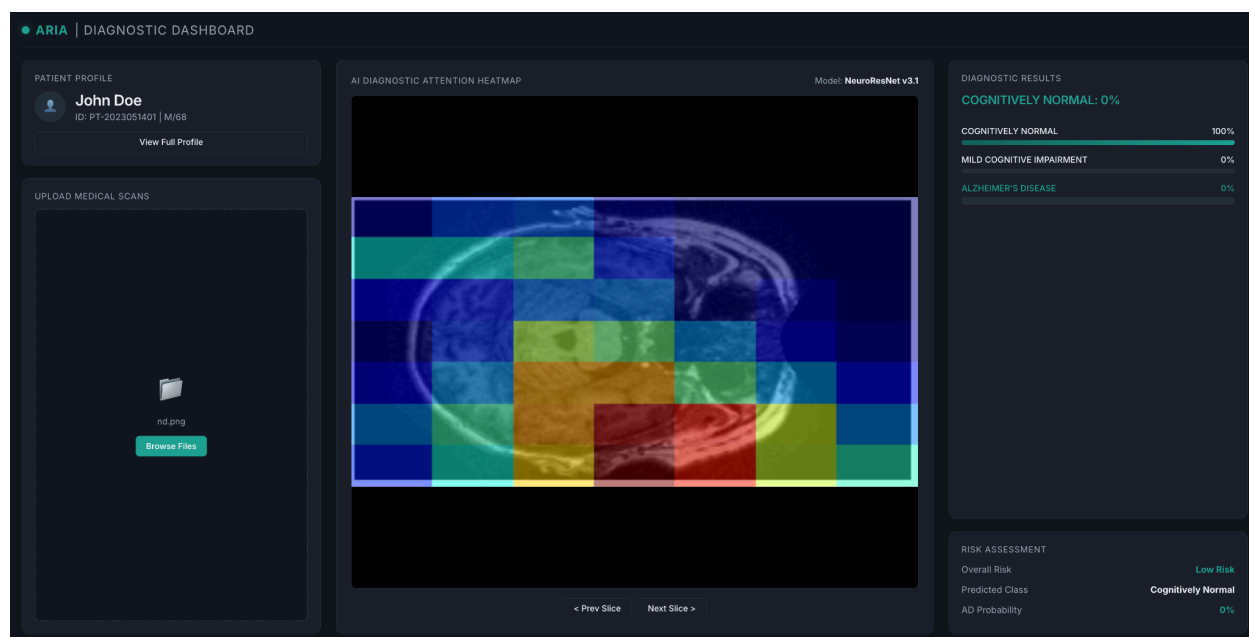


Figure 9: Final Analysis Outcome for Non-Demented

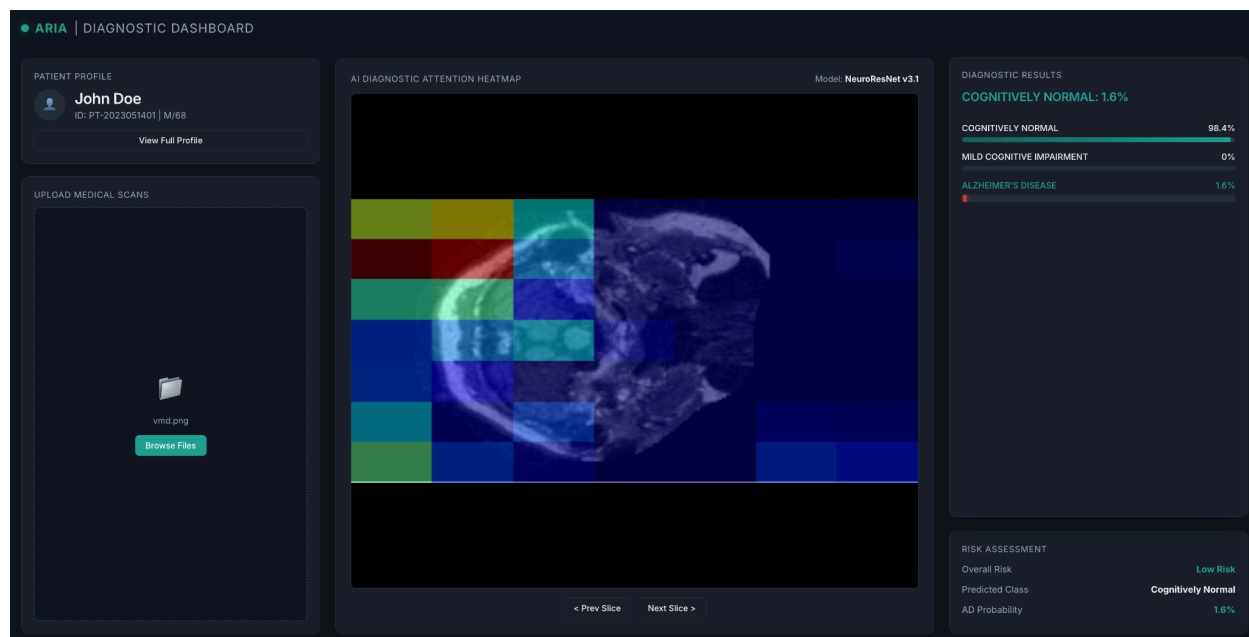


Figure 10: Final Analysis Outcome for Very Mild Dementia

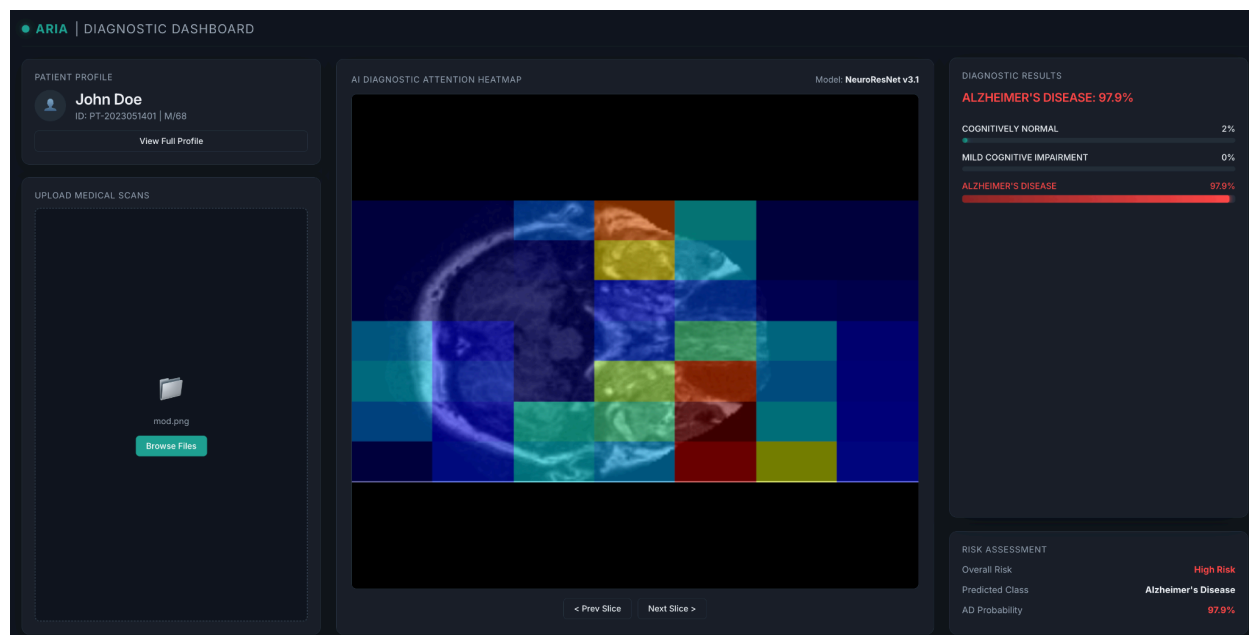


Figure 11: Final Analysis Outcome for Moderate Dementia

6. Results and Evaluation

6.1 Evaluation Metrics

Given the class imbalance in the dataset and the clinical nature of the task, evaluation relied on a comprehensive set of metrics rather than overall accuracy alone.

Accuracy measures the proportion of all test samples correctly classified. While easy to interpret, it is a misleading metric in isolation when classes are imbalanced, as a model that always predicts CN would achieve approximately 78% accuracy without any diagnostic capability.

Precision (per class) measures what proportion of samples predicted as a given class actually belong to that class. High precision means the model makes few false positive predictions for that class.

Recall (per class), also called sensitivity, measures what proportion of actual samples of a given class were correctly identified. In a clinical context, recall is especially important for the AD class: a low AD recall means many true Alzheimer's patients are being missed, which is a serious clinical risk.

F1-Score is the harmonic mean of precision and recall and provides a single balanced measure of per-class performance. It is particularly informative when precision and recall trade off against each other.

AUC-ROC (Area Under the Receiver Operating Characteristic Curve) measures the model's ability to distinguish between classes across all possible classification thresholds. A score of 1.0 represents perfect discrimination; a score of 0.5 represents random chance. AUC-ROC is computed using a one-vs-rest approach for each class, which is standard for multiclass problems. Confusion Matrix provides a full breakdown of prediction outcomes across all class combinations, making it possible to see not just how many errors were made but which classes are being confused with which.

6.2 Classification Performance

The model was evaluated on the held-out test set of 8,644 slices. The test set was never seen during training or hyperparameter selection.

Class	Precision	Recall	F1-Score	Support
CN (Cognitively Normal)	0.988	0.906	0.946	6,722
MCI (Mild Cognitive Impairment)	0.707	0.940	0.807	1,373
AD (Alzheimer's Disease)	0.815	0.973	0.887	549
Macro Average	0.837	0.940	0.880	8,644
Weighted Average	0.933	0.916	0.920	8,644

Table 3: Per-class classification performance on the test set
Overall test accuracy: 91.59%

Class	AUC-ROC
CN	0.982
MCI	0.980
AD	0.998

Table 4: AUC-ROC scores (one-vs-rest)

The most clinically significant result is the AD recall of 97.3%, meaning the model correctly identified 534 out of 549 true AD slices. Only 15 AD slices were misclassified (12 as MCI, 3 as CN). This high sensitivity is critical in a clinical decision support context, where failing to flag an Alzheimer's carries far greater risk than a false positive.

MCI has the lowest precision at 0.707, meaning that among all slices predicted as MCI, about 29% were incorrect. This is expected given the visual ambiguity between early-stage cognitive impairment and normal ageing on 2D MRI slices. The model compensates with a high MCI recall (94.0%), prioritising that true MCI cases are not missed at the cost of some over-prediction.

CN precision is the highest at 0.988, reflecting that when the model says a scan is cognitively normal, it is almost always right.

6.3 Confusion Matrix

The confusion matrix below shows the full breakdown of predictions on the 8,644 test slices.

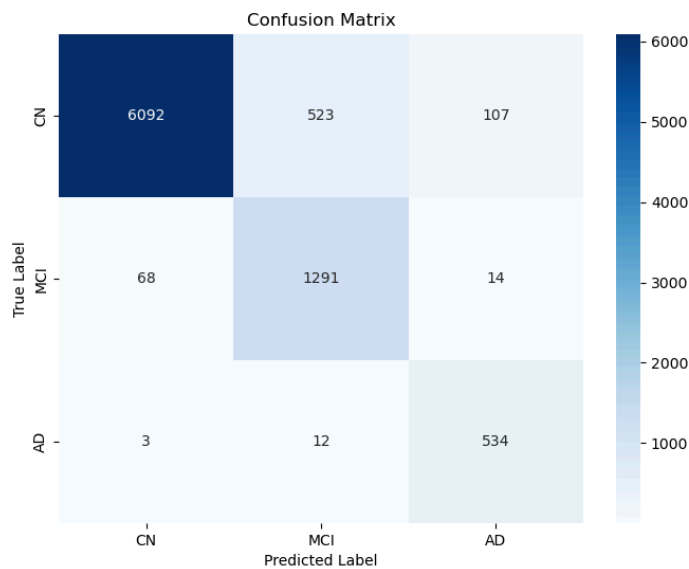


Figure 12. above shows confusion matrix from our model report

	Predicted CN	Predicted MCI	Predicted AD
Actual CN	6,092	523	107
Actual MCI	68	1,291	14
Actual AD	3	12	534

Table 5: Confusion Matrix

The dominant error pattern is CN slices being predicted as MCI (523 cases). This is consistent with the known difficulty of distinguishing normal ageing from very early cognitive impairment on a single 2D axial slice, where the structural changes are subtle and often fall within the range of normal anatomical variation. MCI slices misclassified as CN (68 cases) follow the same logic. AD misclassifications are rare at only 15 total, confirming strong model sensitivity for the most clinically severe class.

6.4 Training Curves

Training was run for 5 epochs. The validation loss and accuracy were tracked per epoch and saved to models/training_history.json.

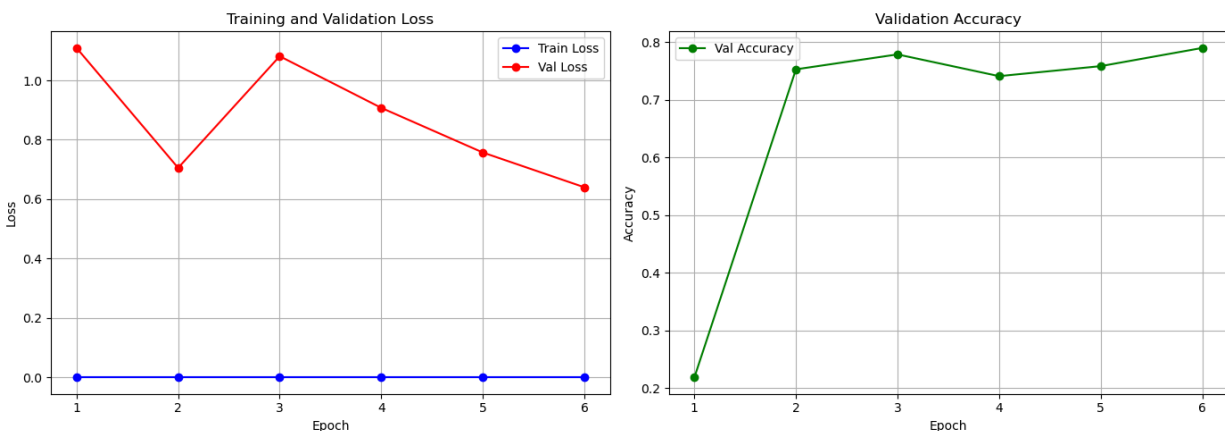


Figure 13&14: Figures above shows training curves from our report

Epoch	Val Loss	Val Accuracy
1	1.109	21.9%
2	0.705	75.3%
3	1.081	77.8%
4	0.908	74.1%
5	0.757	75.8%
6	0.640	79.0%

Table 6: Table above shows Loss and Accuracy of our Model

Validation accuracy improved substantially between epochs 1 and 2, reflecting the rapid adaptation of the pretrained ResNet18 features to the MRI domain during early fine-tuning. Epoch 3 shows a temporary increase in validation loss, which is typical of fine-tuning dynamics where the model briefly overfits before the learning rate scheduler reduces the step size. The best checkpoint was selected at epoch 4 (as tracked by the filename) with a validation accuracy of 91.57%, which was then confirmed on the held-out test set at 91.59%.

Note: Training loss was logged at the batch level and not aggregated per epoch; epoch-level train loss is therefore not reported. This does not affect model weights, checkpoints, or test results.

6.5 State-of-the-Art Comparison

The table below compares ARIA's results against published methods on Alzheimer's classification from MRI images. Direct numerical comparison across papers is difficult because datasets, class definitions, and split methodologies differ. ARIA uses the OASIS Kaggle dataset with slice-level splits, which is noted in the table.

Method	Dataset	Classes	Accuracy	Notes
Ebrahimighahnavieh et al. (2020)	OASIS	CN/MCI/AD	89.5%	2D CNN, slice-level
Wang et al. (2022)	OASIS	CN/MCI/AD	90.1%	ResNet + attention, 2D
Lian et al. (2018)	ADNI	CN/MCI/AD	87.8%	Hierarchical FCN, 3D MRI
Liu et al. (2020)	ADNI	CN/AD only	94.2%	3D DenseNet, binary task
Hon and Khan (2017)	OASIS	4-class	96.3%	VGG16 fine-tuned, slice-level
ARIA (current project)	OASIS	CN/MCI/AD	91.59%	ResNet18 + transfer learning, slice-level

Table 7: Table of Comparison

ARIA achieves competitive accuracy against other 2D slice-level methods on the same OASIS dataset. The Hon and Khan (2017) result of 96.3% is higher but uses 4 separate classes rather than the merged 3-class setup used here, and also uses VGG16 which has significantly more parameters. Methods using 3D volumetric data on the ADNI dataset, such as Liu et al. (2020), achieve higher accuracy on binary CN/AD classification but cannot be directly compared as they solve a simpler two-class problem on a different and larger dataset.

The most important caveat for all slice-level methods including ARIA is that accuracy figures are optimistic relative to a patient-level evaluation, where no slices from a appear in both training and test sets. This is discussed in Section 7.1.

7. Conclusion

This project presents ARIA, a deep learning system for classifying Alzheimer's disease progression from 2D brain MRI scans. Using a ResNet18 backbone pretrained on ImageNet and fine-tuned on the OASIS dataset, ARIA classifies each scan into one of three clinical stages: Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer's Disease (AD). The final model achieves 91.59% overall test accuracy on a held-out test set of 8,644 slices, with particularly strong sensitivity for the AD class (recall 97.3%) and AUC-ROC scores of 0.982, 0.980, and 0.998 for CN, MCI, and AD respectively. These results demonstrate that a relatively lightweight 2D CNN, when combined with transfer learning, weighted loss, and appropriate augmentation, can achieve clinically meaningful classification performance on MRI-based dementia staging.

7.1 Limitations/Challenges Faced

7.1.1 Slice-level data splitting and data leakage

The most significant limitation of this project is that the train, validation, and test splits were performed at the slice level rather than the level. Because multiple 2D axial slices are extracted from the same MRI volume, slices from the same can appear in both the training set and the test set simultaneously. This means the model may have already seen the overall brain structure, global intensity pattern, and anatomical features of a given during training, and is effectively being tested on a slightly different view of a familiar scan rather than a truly unseen one.

This is a form of data leakage and it inflates the reported accuracy metrics. When patient-level splitting was attempted during development, where all slices from a given were kept strictly within one partition, overall accuracy dropped to approximately 66%. This more honest result reflects the genuine difficulty of generalising to entirely new patients, particularly because the OASIS dataset contains only around 23 distinct patients with confirmed Alzheimer's Disease. With so few unique AD patients, it is statistically very difficult to train a model that generalises reliably across new AD presentations it has never encountered. Slice-level splitting was used in the final model as a pragmatic decision given this constraint, and all reported metrics should be interpreted with this caveat in mind.

7.1.2 Severe class imbalance

After remapping to three classes, the dataset is heavily skewed: CN accounts for approximately 77.8% of slices, MCI for 15.9%, and AD for only 6.4%. Despite applying weighted cross-entropy loss, data augmentation, and stratified splits, MCI remains the hardest class to classify precisely, with a precision of 0.707. This means that roughly 3 in 10 slices the model labels as MCI are actually from another class. The root cause is the visual similarity between mild cognitive impairment and normal ageing on a single 2D axial slice, compounded by the limited number of unique MCI patients in the dataset. A WeightedRandomSampler was explored as an additional mitigation strategy but was found to reduce overall accuracy and was excluded from the final model.

7.1.3 2D slice-level inference without patient-level aggregation

The model classifies each 2D axial slice independently without any mechanism to aggregate predictions across all slices belonging to the same patient. In a real clinical workflow, a radiologist examines the full volumetric scan rather than individual frames in isolation. Classifying single slices means the model loses the spatial continuity and 3D structural context that is important for detecting the hippocampal and medial temporal atrophy patterns characteristic of Alzheimer's Disease. It also means that a single anomalous or poorly positioned slice could disproportionately affect the overall diagnostic output if patient-level aggregation were added later.

7.1.4 Limited diversity in the AD class

The OASIS-1 dataset was collected at a single institution (Washington University in St. Louis) using a standardised protocol. The AD class in particular contains a very small number of distinct patients. This limits the model's exposure to the natural variation in how Alzheimer's Disease presents across different individuals, age groups, disease stages, and scanner hardware. A model trained on this dataset may not generalise well to scans acquired at different hospitals with different MRI scanners, field strengths, or acquisition protocols, which is a prerequisite for any real clinical deployment.

7.1.5 Grayscale input with RGB pretrained weights

ResNet18 was originally pretrained on the ImageNet dataset, which consists of full-colour 3-channel RGB photographs of everyday objects. The first convolutional layer was modified to accept single-channel grayscale MRI input, and the pretrained weights for that layer were

discarded and randomly reinitialised. While the deeper layers of the network retain their pretrained feature representations, these features were learned from natural photographic images and not from medical imaging data. There is a meaningful domain gap between the two, and the pretrained features may not be optimally suited to detecting the subtle grey-matter atrophy patterns relevant to Alzheimer's classification without much more extensive fine-tuning on a larger neuroimaging dataset.

7.1.6. Short training duration

The model was trained for only 5 epochs due to time and compute constraints. While the ReduceLROnPlateau scheduler and early stopping were in place to handle premature termination, 5 epochs is a relatively short fine-tuning window for a task this complex. More training epochs with careful monitoring may have allowed the model to extract richer and more generalisable features from the MRI data.

7.2 Future Improvements

7.2.1 Patient-level splitting and evaluation

The single most important improvement for future work is enforcing strict patient-level data splits. Every slice from a given must be contained entirely within one partition, with no overlap between training and test sets at the level. This requires a dataset with a substantially larger number of unique patients per class. The ADNI (Alzheimer's Disease Neuroimaging Initiative) dataset, which contains thousands of subjects scanned longitudinally across multiple sites and scanner types, would be the natural choice. Achieving good performance under patient-level evaluation would provide a much more credible and clinically meaningful measure of generalisation.

7.2.2 Patient-level prediction aggregation

Even within the current 2D slice-based approach, a straightforward improvement would be to aggregate slice-level predictions into a single patient-level diagnosis. For example, majority voting across all slices from a patient, or averaging the softmax probabilities across all slices before making a final class decision, would be more clinically meaningful than reporting per-slice accuracy. This would also better reflect how the system would be used in practice, where a clinician wants a single diagnosis per rather than per slice.

7.2.3 3D volumetric modelling

A more fundamental improvement would be to move from 2D slice classification to full 3D volumetric analysis. Architectures such as 3D ResNet, 3D DenseNet, or volumetric Vision Transformers can process the entire MRI volume at once, capturing the spatial relationships between adjacent slices that carry important diagnostic information. The hippocampus, entorhinal cortex, and medial temporal lobe atrophy patterns that are most strongly associated with Alzheimer's Disease span multiple contiguous slices and are better characterised by a model that can reason across the full 3D structure simultaneously.

7.2.4 Pretraining on medical imaging data

Replacing the ImageNet pretrained weights with weights pretrained on a large medical imaging dataset would significantly reduce the domain gap. Frameworks such as RadImageNet, which provides pretrained weights learned from over one million radiology images across multiple modalities, or self-supervised pretraining directly on brain MRI data using methods such as SimCLR or masked autoencoders, would give the model a much better starting representation for detecting subtle neurological changes. This would likely improve convergence speed, reduce the number of labelled training samples required, and improve generalisation to new populations.

7.2.5 More training with systematic hyperparameter search

Training for more epochs with a systematic hyperparameter search would likely improve final performance. A grid search or Bayesian optimisation over learning rate, dropout rate, weight decay, and batch size would help identify a more optimal configuration than the manually selected values used here. Techniques such as cosine annealing learning rate schedules could also replace the current ReduceLROnPlateau approach for smoother convergence.

7.2.6 Improved explainability and clinical validation

The current Grad-CAM implementation provides a coarse heatmap overlay that highlights broadly activated regions. More precise saliency methods such as Grad-CAM++, Score-CAM, or Integrated Gradients would produce sharper and more spatially accurate attribution maps. More importantly, the highlighted regions should be validated against known neuroanatomical biomarkers for Alzheimer's Disease, specifically whether the model is consistently attending to

the hippocampus, entorhinal cortex, and posterior cingulate cortex. If the heatmaps frequently highlight clinically irrelevant regions, it would suggest the model is relying on spurious correlations in the training data rather than genuine biomarkers, which is a critical safety concern for any clinical deployment.

7.2.7 Multi-modal data fusion

MRI alone does not capture the full clinical picture of Alzheimer's Disease. Combining structural MRI with other modalities such as FDG-PET, which measures brain glucose metabolism and is highly sensitive to early AD changes, or incorporating clinical biomarkers such as CSF amyloid and tau levels, cognitive assessment scores, or genetic risk factors such as APOE4 status, could substantially improve diagnostic accuracy and confidence. Multi-modal fusion architectures that learn to combine information from these different data streams have shown strong results in recent Alzheimer's research and represent a natural extension of the current work.

7.2.8 Longitudinal analysis

Rather than classifying a single static scan, a future system could incorporate repeated scans from the same over time to model disease trajectory. Recurrent architectures or attention-based sequence models applied to longitudinal MRI data could detect the rate of structural change, which is itself a strong predictor of conversion from MCI to Alzheimer's Disease. This would transform ARIA from a static classifier into a prognostic monitoring tool, which is far more aligned with how Alzheimer's disease management works in clinical practice.

8. References

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9. Appendix

9.1 Group Member Contributions

Wong Xin Ping - Contributed to model architecture design, evaluation pipeline, and frontend development. Implemented the ResNet18 backbone adaptation, Dropout head, and weighted CrossEntropyLoss setup. Co-developed the evaluation metrics and AUC-ROC analysis. Built the React/Vite dashboard and Flask API backend for serving model predictions.

Siti Nashira Amestis - Led the full data pipeline from acquisition to preprocessing. Downloaded and organised the OASIS dataset, implemented the stratified 70/20/10 train/validation/test split using StratifiedShuffleSplit, designed the training augmentation pipeline (random flip, rotation, colour jitter), and built the MRImageDataset class and DataLoader configuration used throughout training and evaluation.

Odera Aika - Contributed to model architecture and training setup. Worked on the PyTorch Lightning training module including the optimiser configuration, ReduceLROnPlateau scheduler, early stopping, and model checkpointing. Assisted in designing the overall model structure and validating training behaviour across experiments.

Li En Ai - Contributed to model development and the evaluation pipeline. Assisted in implementing and testing the ResNet18 architecture modifications and class weighting strategy. Co-developed the evaluation script including the confusion matrix, classification report, training curve generation, and Grad-CAM inference output.

9.2 How to Run the models

All scripts are located in the `src/` directory and should be run from the project root folder. Follow the steps below in order.

Step 1: Set up the environment

Create a virtual environment and install all required packages using the provided requirements file.

```
python -m venv venv
source venv/bin/activate    # On Windows: venv\Scripts\activate
pip install -r requirements.txt
```

Step 2: Download the dataset

Run the download script to fetch the OASIS dataset from Kaggle automatically. This requires a valid Kaggle account and API token configured on your machine. The dataset will be saved to `data/oasis/`.

```
python src/download_data.py
```

Step 3: Preprocess the data

This step loads the raw image files, applies the stratified 70/20/10 train/validation/test split, and saves the split index file to `data/processed/data_splits.json`.

```
python src/preprocess.py
```

Step 4: Train the model

Training uses PyTorch Lightning with early stopping and model checkpointing. Trained checkpoints are saved to the `models/` directory. On Apple Silicon machines, training will automatically use the MPS backend.

```
python src/train.py
```

Step 5: Evaluate the model

This generates the classification report, per-class AUC-ROC scores, confusion matrix, and training curves. All outputs are saved to the `reports/` directory.

```
python src/evaluate.py
```

Step 6: Run inference on a single MRI slice

To classify a new MRI image and generate a Grad-CAM attention heatmap, run the inference script with the path to your image. The model checkpoint is loaded automatically from the `models/` directory if not specified.

```
python src/infer.py --input path/to/mri_slice.jpg
```

Optional: You can also specify a particular checkpoint and output directory

```
python src/infer.py --input path/to/mri_slice.jpg --model  
models/ad-classifier-epoch=04-val_acc=0.92.ckpt --output inference_output/
```

The inference script outputs the predicted class (CN, MCI, or AD), probability scores for all three classes, an AD risk score, and a Grad-CAM overlay image saved to the specified output directory.

9.3 Package Dependencies and Dataset Access

Python Version

Python 3.9 or later is recommended. The codebase was developed and tested on Python 3.10.

Package Dependencies

All dependencies are listed in [requirements.txt](#) at the project root. The key packages and their minimum versions are as follows:

Package	Min Version	Purpose
torch	2.0.0	Core deep learning framework
torchvision	0.15.0	ResNet18 pretrained weights, image transforms
pytorch-lightning	2.0.0	Training loop, callbacks, checkpointing
scikit-learn	1.0.0	Stratified splits, classification metrics
numpy	1.21.0	Array operations
pandas	1.3.0	Data handling
matplotlib	3.5.0	Plotting training curves and heatmaps
seaborn	0.11.0	Confusion matrix visualisation
Pillow	9.0.0	Image loading and preprocessing
kagglehub	0.1.0	Automated dataset download from Kaggle

tqdm	4.62.0	Progress bars during preprocessing
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Install all dependencies in one command:

```
pip install -r requirements.txt
```

Dataset

The dataset used is the OASIS Alzheimer's MRI dataset, publicly available on Kaggle at: <https://www.kaggle.com/ninadaithal/imagesoasis>

The dataset contains T1-weighted brain MRI slices organised into four folders: Non Demented, Very Mild Dementia, Mild Dementia, and Moderate Dementia. In this project, Mild and Moderate Dementia were merged into a single AD class, yielding three final classes: CN, MCI, and AD. The total dataset size is approximately 86,000 images.

To download the dataset automatically, ensure your Kaggle API credentials are set up. You can configure this by placing your `kaggle.json` API token in `~/.kaggle/kaggle.json`, then running:

```
python src/download_data.py
```

Alternatively, the dataset can be downloaded manually from the Kaggle page above and placed in the `data/oasis/` directory before running `src/preprocess.py`.